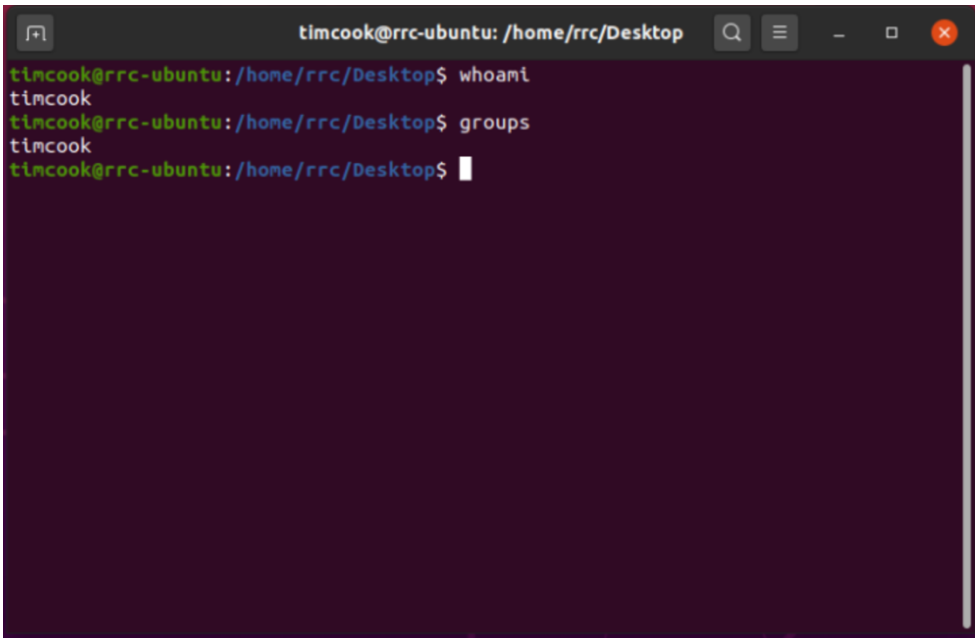
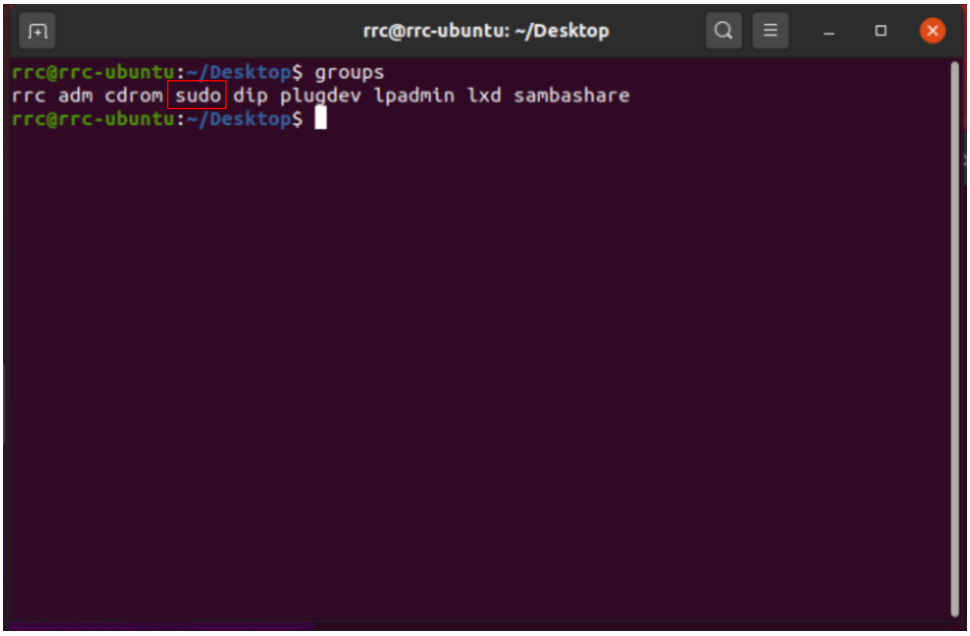
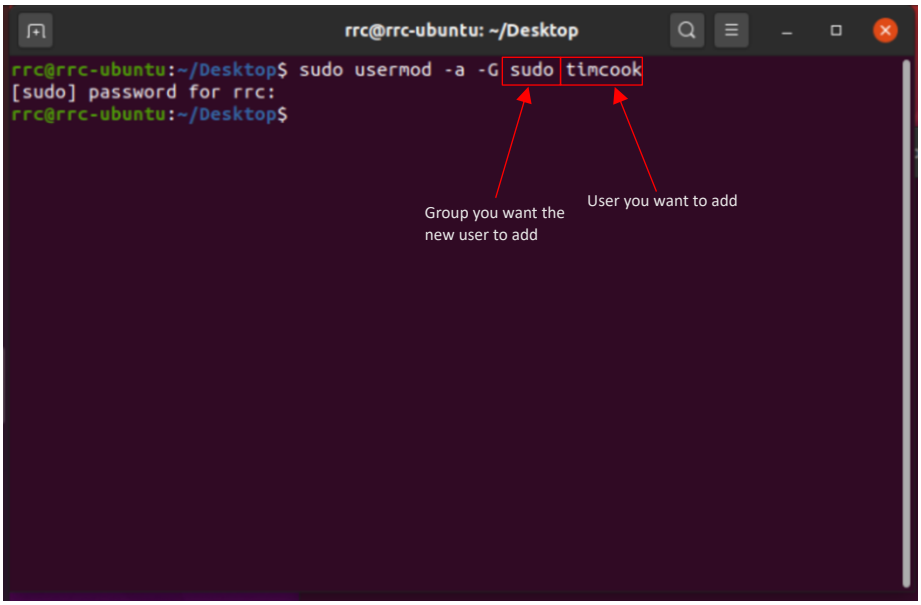
1. Create a new user and log into the new account. You can use the “whoami” command to see which user you are logged in as. When you type the “group” command, you will see your username as your group since Ubuntu creates a new group with your username when you create a new user.



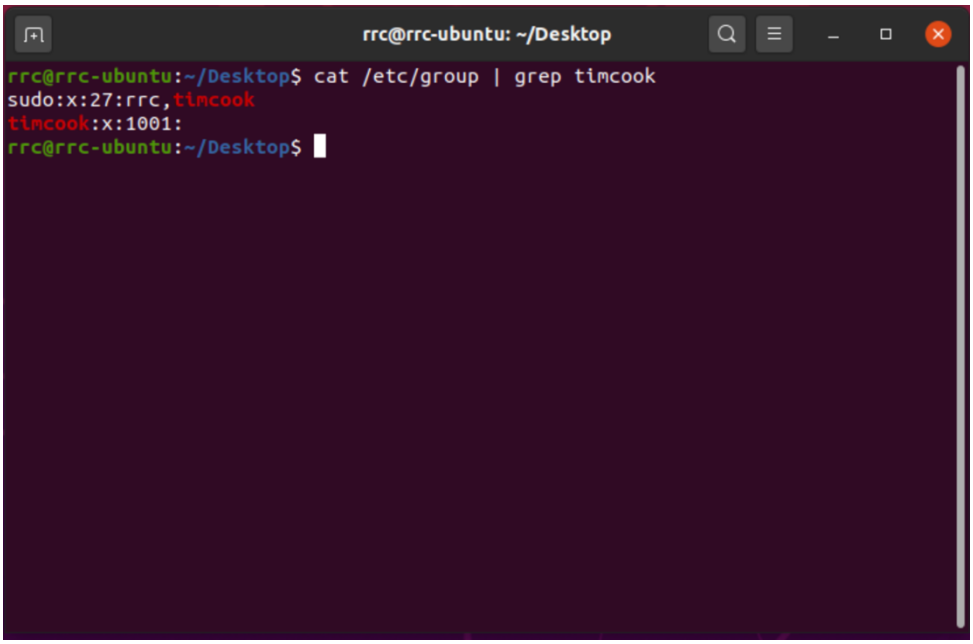
2. When you run the “group” command to your rrc user, you will see a lot of groups you are in. One of them is named “sudo” which enables the rrc user to use the “sudo” command.



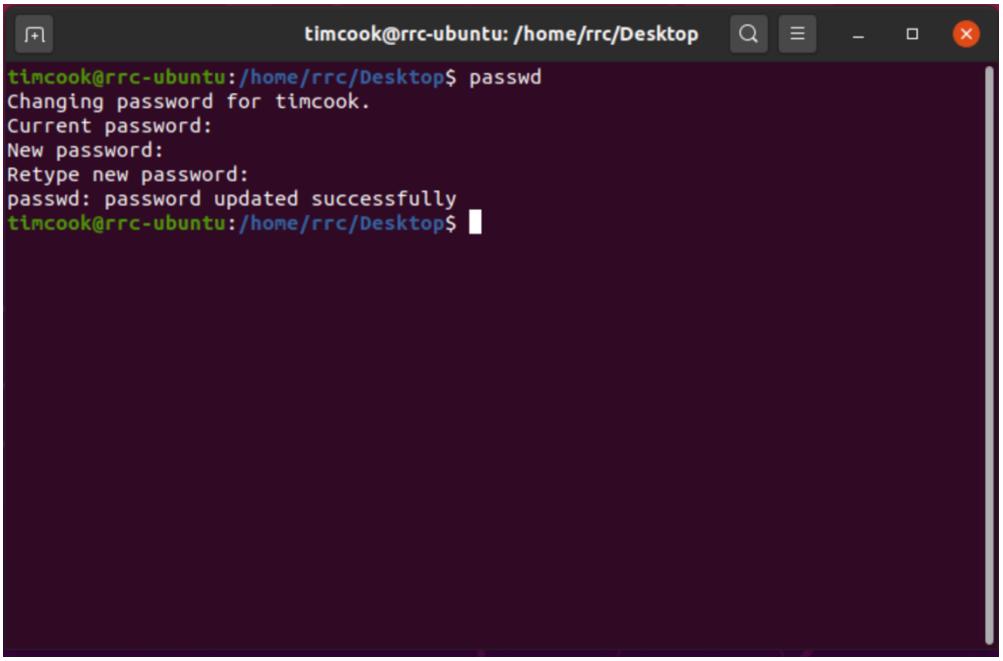
3. You can give access to the new user to the “sudo” access by adding him to the sudo group. You will need to run this command as the rrc user who has “sudo” access. The command to use here is the “usermod” command. You will need “-a” for an additional group. If not, it will remove the new user from the current group. You will also need “-G” to mention what you would like him to add is a group. The new user has to log out and log in back again in order to be able to use the “sudo” command.



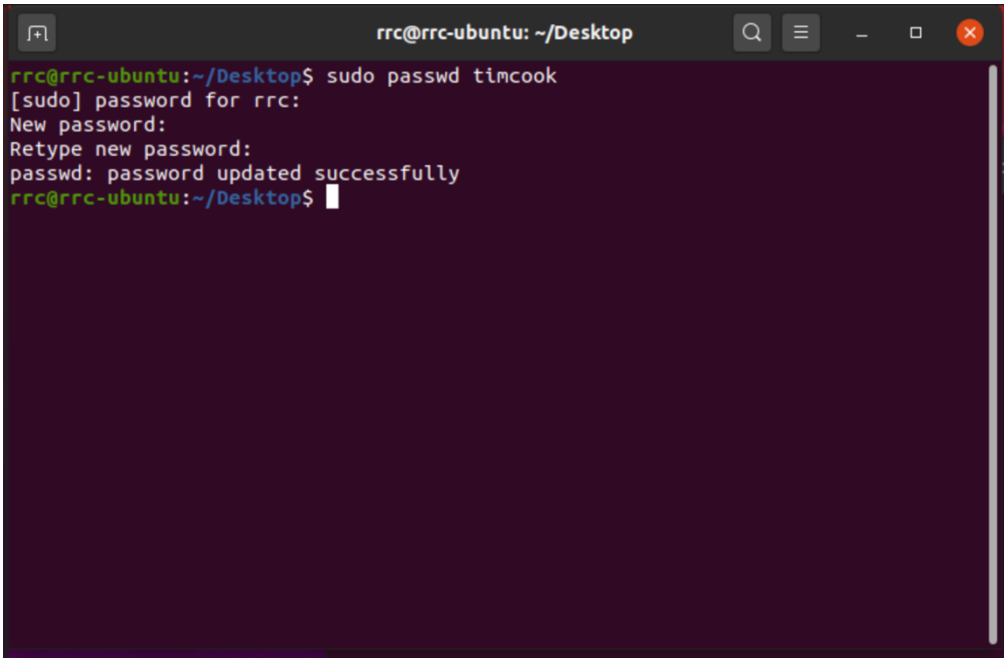
4. You can send one command’s result to another command by using “|” and then typing the other command you want the first command’s result to put into.



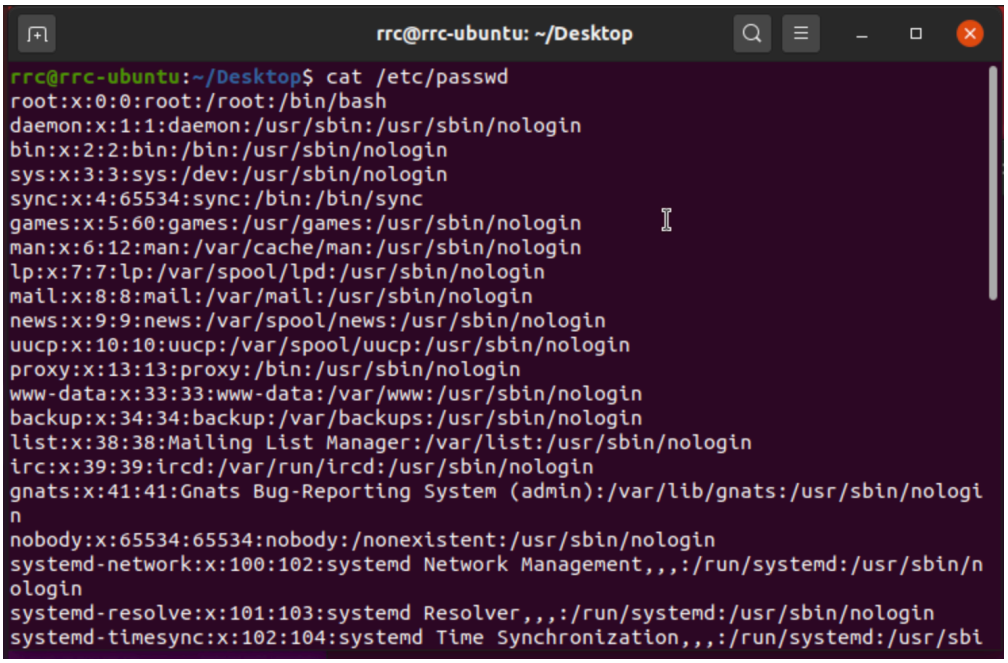
5. You can change your account's password using the "passwd" command.



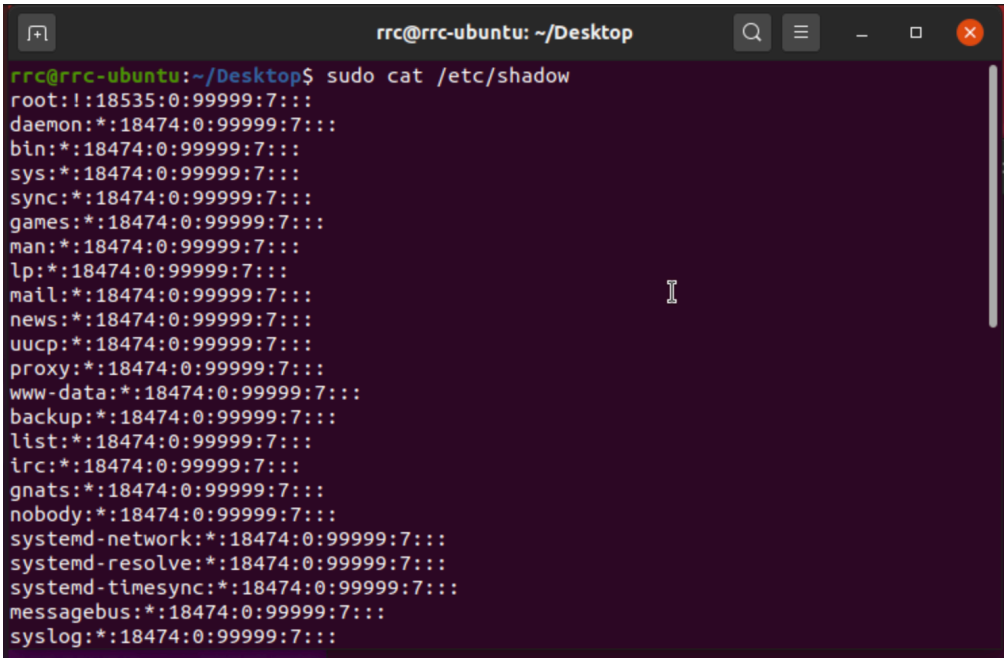
6. You can also change someone else's account using the "sudo passwd" command.



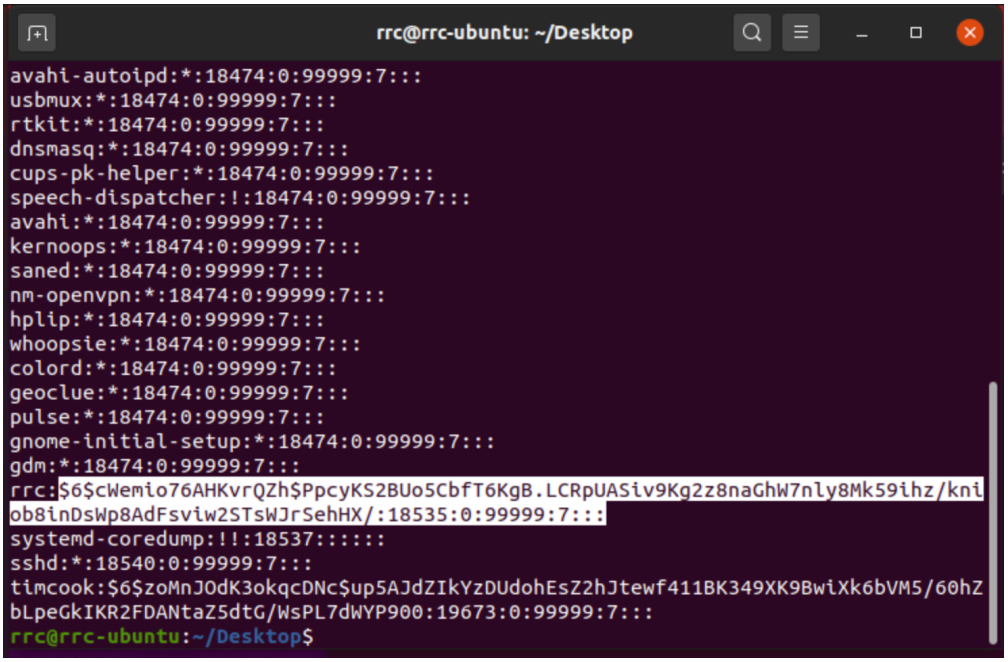
7. You can view the passwd file by simply using the “cat” command. But this passwd file doesn’t store any information about users’ passwords.



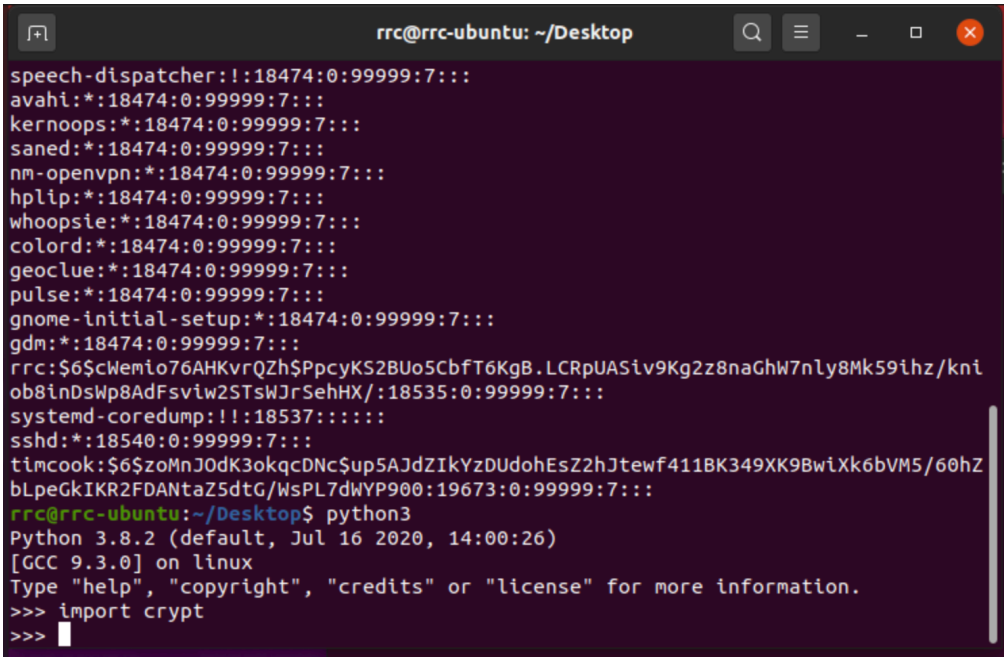
8. However, there is a file named “shadow” inside “/etc” which you need sudo permission to view since it stores the password information.



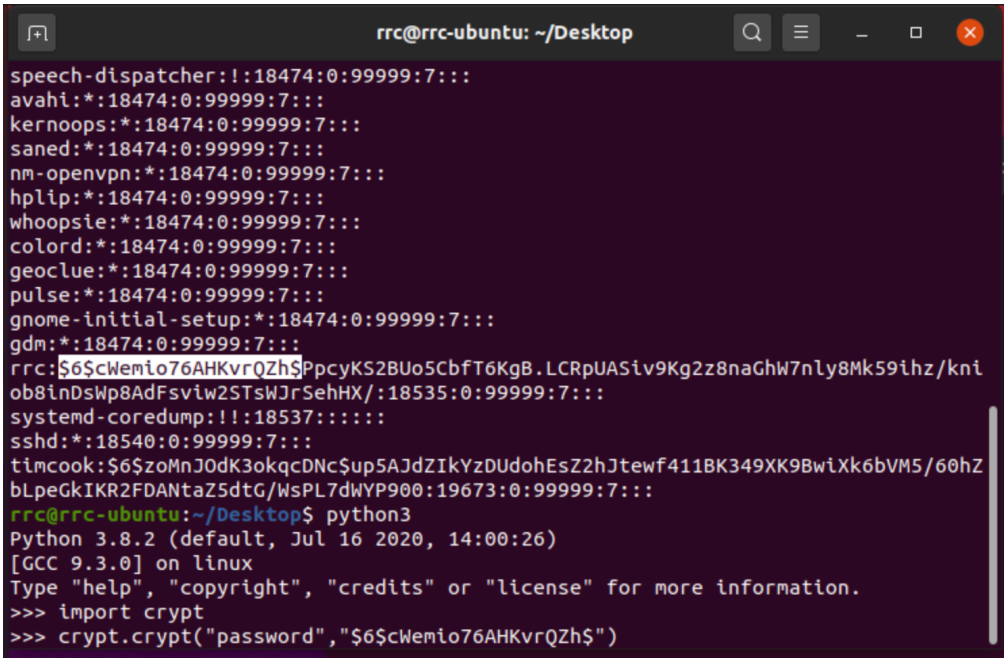
9. As you can see the computer stores our password information in other forms. Your computer does not know your password. When you type your password, it changes into this form and the computer will only unlock if it matches these random words.



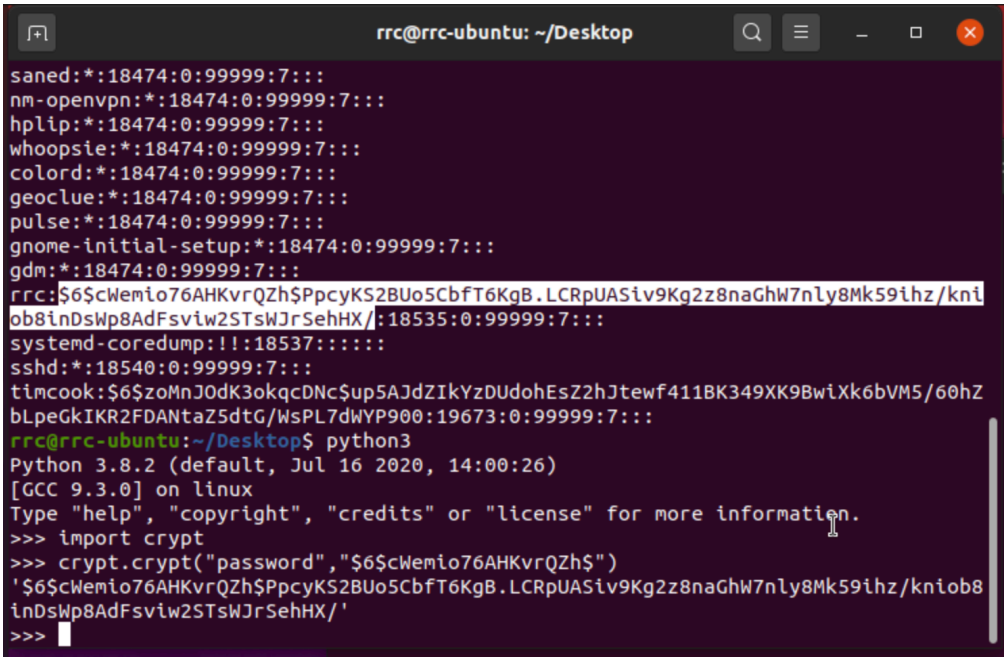
10. You can use the Python language to work on guessing the password. Just simply type “python3” then “import crypt”.



11. You can now type crypt.crypt as below and then type the password you think might be and then type the higilighted text from the first $ to the third $ in the format shown below.



12. It will generate the following random words that match the highlighted words meaning your password guess is correct. In this case, the password is just “password”.



13. You can see that if you try to put an upper case letter, the outputs will be completely different even just one letter.

